Summary in Graph

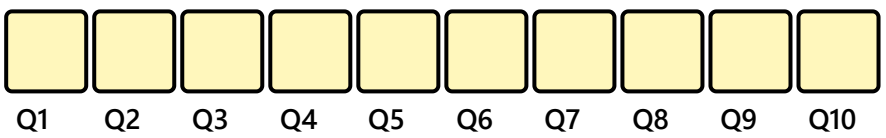
Exam Summary (GO Classes GATE CS | Data Structures | Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	10 (0 + 10)
Total Marks:	20 (0 + 20)
Exam Duration:	30 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (37) out of 370 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 2 Penalty: 0 Programming and DS

Consider a problem: Given an array, check whether the array is in sorted order or not. Suppose both iterative and recursive approaches are used to solve this problem.

Which of the following statements is/are correct?

A. Recursive approach gives space complexity as $O(n)$ for recursive stack space.
B. Iterative approach gives Auxiliary space as $\Theta(n)$.

C. Time Complexity for recursive approach is $O(n)$.
D. Time Complexity for iterative approach is $\Theta(n \log n)$.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Rank the following functions by increasing order of growth. That is, find any arrangement $g1, g2, g3, g4$ of the functions satisfying $g1 = O(g2), g2 = O(g3), g3 = \mathrm{O}(\mathrm{g}4)$.

$$f_1(n) = 2^{(\log n)^4} \qquad f_2(n) = \pi^n f_3(n) = n^{5(\log n)^2} \qquad f_4(n) = \sqrt{2^{\sqrt{n}}}$$

A. $f1, f3, f4, f2$
B. $f3, f1, f4, f2$
C. $f1, f3, f2, f4$
D. $f1, f3, f4, f2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

What will be the time complexity of $\mathrm{function}()$?

```
function(int n) {
    for (int i = 0; i < n; i++) {
        for (int j = i; j < i * i; j++) {
            if (j % i == 0) {
                for (int k = 0; k < j; k++) {
                    printf("*");
                }
            }
        }
    }
}
```

A. $\Theta\left(n^3\right)$
B. $\Theta\left(n^4\right)$
C. $\Theta\left(n^5\right)$
D. $\Theta\left(n^2 \log n\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Determine the complexity of the following code, using "Θ" as indicated.

```
int sum = 0;
for (int i = 1; i <= n; i *= 2)
    for (int j = 1; j <= i; j++)
        sum++;
        print(sum);
```

A. $\Theta\left(n^2\right)$
B. $\Theta\left(n^2 \log n\right)$
C. $\Theta(n)$

D. None of these

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 02sec | Discuss

Q #5 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider functions $f()$ and $g()$ which are positive increasing functions.

Which of the following(s) is/are correct?

A. If $f(n) = o(g(n))$ then $\log(f(n)) = o(\log(g(n))$, here o is small-oh
B. If $f(n) = \mathrm{O}(g(n))$ then $e^{f(n)} = \mathrm{O}\left(e^{g(n)}\right)$, here O is big-oh
C. If $f(n) = O(g(n))$ then $\log(f(n)) = O(\log(g(n))$, here O is big-oh
D. If $f(n) = o(g(n))$ then $e^{f(n)} = o\left(e^{g(n)}\right)$, here o is small-oh

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following(s) is/are correct?

A. If $f(n) = n \log n, g(n) = \sum_{i=1}^{n} \frac{n}{i}$ then $f(n) = \Theta(g(n))$
B. If $f(n) = 1.01^n, g(n) = n^{1000}$ then $f(n) = \Omega(g(n))$
C. If $f(n) = \sum_{i=1}^{n} i^2, g(n) = n^3 + \sum_{i=1}^{n} i$ then $f(n) = \Theta(g(n))$
D. If $f(n) = n!, g(n) = 3^n \times 5^n$ then $f(n) = \Omega(g(n))$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Which of the following statement is FALSE?

A. If $f(n) = \mathcal{O}(g(n))$ and $f(n) = \Omega(g(n))$, then we have $(f(n))^2 = \Theta\left((g(n))^2\right)$
B. If $f(n) = \mathcal{O}(g(n))$ and $f(n) = \Omega(g(n))$, then we have $f(n) = g(n)$
C. $2^n + n^2 = \mathcal{O}\left(3^n\right)$
D. $2^n + n^2 = \mathcal{O}\left(2^n\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Assume you have two positive functions f and g such that $f(n)$ is in $O(g(n))$. For each of the following statements, decide which one(s) is/are always TRUE.

A. $2^{f(n)}$ is $O\left(2^{g(n)}\right)$
B. $f(n)^2$ is $O\left(g(n)^2\right)$
C. $f(n) = O\left((f(n))^2\right)$
D. $g(n) = \Omega(g(n))$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Let $S_1 = \sum_{r=0}^{\log n - 1} \frac{nr}{2^r}$, and $S_2 = \sum_{r=0}^{\log n - 1} r2^r$.

Which of the following is true?

A. $S_1 = \Theta(n \log n), S_2 = \Theta(n \log n)$
B. $S_1 = \Theta(n), S_2 = \Theta(n \log n)$
C. $S_1 = \Theta(n \log n), S_2 = \Theta(n)$
D. $S_1 = \Theta(n), S_2 = \Theta(n)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Match Column A with Column B. Column B represents asymptotic time complexity of loops in Column A

Column A	Column B
a) for (i = 1; i*i <= N; i = 2*i) ;	i) $\Theta(\log N)$
b) for (i = 1; i <= N; i = 2*i) for(j = 1; j <= i; j = j + 1);	ii) $\Theta(N)$
c) for (i = 1; i*i <= N; i = i + 1) for(j = 1; j <= i; j = j + 1);	iii) $\Theta(\sqrt{N} \log N)$
d) for (i = 1; i*i <= N; i = i + 1) for (j = 1; j <= i; j = 2*j) ;	iv) $\Theta(N \log N)$

A. a-i, b-ii, c-iii, d-iv
B. a-i, b-iv, c-ii, d-iii
C. a-i, b-ii, c-ii, d-iii
D. None of these

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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